OMEGA ∞ EXECUTIVE INTELLIGENCE REPORT

JobFunnel Multi-Board Deduplication & Master Funnel

Automated Ingestion, Exact Key De-duping, TF-IDF Cosine Similarity & Master CSV Telemetry across 8,576 Role Postings

EXECUTIVE SUMMARY: JobFunnel pipeline ingested 8,576 candidate listings from 4 repositories, purged 1,666 duplicate multi-board listings (515 exact, 1,151 TF-IDF cosine variants), and established a 6,910 clean-opportunity Master CSV Funnel. Master FTS5 Index expanded to 17,657 entities.

1. Executive Funnel Telemetry

Candidate postings from diverse career scrapers and repositories suffer from severe duplicate inflation due to cross-board aggregations. JobFunnel solves this through a deterministic dual-stage filter:

Funnel Metric	Listing Count	Strategic Significance
Raw Listings Ingested	8,576	Aggregate listings across Summer '27, New Grad '26 & '27, and AI Jobs
Exact Metadata Dupes Purged	515	Identical (Company, Title, Location) cross-board duplicates
TF-IDF Cosine Dupes Purged	1,151	Cosine similarity >= 0.88 with company stem matching
Net Master Funnel Listings	6,910	Clean, non-redundant opportunities ready for application targeting

2. Ecosystem Ingestion & Remoteness Breakdown

Source Ecosystem	Clean Retained Roles	Share of Master Funnel
New Grad 2026	2,604	37.7%
Summer 2027 Internships	2,182	31.6%
AI College Jobs 2027	1,594	23.1%
New Grad 2027	530	7.7%

Workplace Flexibility Taxonomy: IN_PERSON: 6,669 • FULLY_REMOTE: 231 • HYBRID: 10

3. JobFunnel Dual-Stage Deduplication Architecture

JobFunnel implements a rigorous two-tier anti-redundancy architecture that safeguards job hunters from application fatigue and duplicative outreach:
1. Stage 1 (Deterministic Tuple Hash): Hashes (company_stem, title_stem, location_stem). All subsequent arrivals with identical tuple signatures are tagged as EXACT_METADATA_MATCH and quarantined.
2. Stage 2 (TF-IDF Cosine Similarity Matrix): For listings with minor stylistic title variance (e.g., 'Software Engineer Intern - Systems' vs 'Software Engineer Intern (Systems)'), JobFunnel fits an n-gram TF-IDF vectorizer over the textual corpus. Cosine distances >= 0.88 combined with company namespace matching are flagged as TFIDF_COSINE_SIMILARITY.
3. Resultant Ledger: All eliminated listings are preserved in 'data/jobfunnel_duplicates.json' with both duplicate ID and originating master ID for complete auditability.

4. Audit Sample: Flagged Multi-Board Duplicates

Company	Role Title	Original vs Duplicate Source	Match Algorithm
Salesforce	Software Engineer Intern	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)
Susquehanna Internationa	Quantitative Research Intern	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)
IMC Trading	Quantitative Research Intern	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)
Heliux	Software Engineer Intern	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)
TikTok	Software Engineer Intern	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)
TikTok	Machine Learning Engineer In	Summer 2027 Internships -> Summer 2027 Internships	EXACT_METADATA_MATCH (1.0)

5. Unified Master Engine Integration

JobFunnel is integrated natively into the OMEGA master operating system:
• Master CSV Pipeline: 'data/jobfunnel_master_list.csv' containing 6,910 deduplicated rows with full status progression.
• Dedicated FTS5 Database: 'data/jobfunnel_master.db' with instant sub-millisecond keyword search.
• Master FTS5 Database: 'data/omega_unified_master.db' total entity index elevated to 17,657 cross-repository records.
• Interactive Dashboard: 'jobfunnel_dashboard.html' providing real-time client-side filtering by Status, Provider, Remoteness, and Search.
• CLI Tooling: 'jobfunnel_engine.py' supporting --stats, --search <query>, and --ui.